

Coded Beam Training

XL-MIMO

$N_t \gg 1024$ Exhaustive BT $\rightarrow O(N_t)$ pilots

hierarchical $\rightarrow O(\log N_t)$

$\rightarrow$ but suffer error propagation

at low SNR

$\rightarrow$ Beam alignment failure (cell edge users)

Each antenna contributes

$\rightarrow$ very narrow beams

$\downarrow$

helps in

$R_b \uparrow$, extend coverage, reduce interference

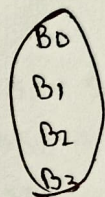
Search problem

BS $\rightarrow$ find which beam direction $\rightarrow$ best UE

$N_t = 1024$

Beamforming CB $\Rightarrow 1024$ beams } UE decides which has strongest power

(Δ) 8 beams



$\rightarrow$ left top

$\rightarrow$ right bottom

Exhaust BT $\rightarrow$ test every beam

$B_1 \rightarrow$ pilot $\rightarrow$ measure power

$B_2 \rightarrow$ " " "

!

$B_{1024} \rightarrow$ pilot $\rightarrow$ meas power

Pilot overhead

$O(N_t)$

very slow

$\wedge$ inefficient

6Gx

$B_0 B_1 \quad \underline{\underline{B_2 B_3}}$
✓

$B_2 \quad \underline{\underline{B_3}}$
✓

③ stop ✓

Error propagation

low SNR

cell edge

$S_1 \rightarrow w \times$

$S_2 \rightarrow w \times$

$S_3 \rightarrow w \times$

final beam

$\downarrow$
wrong

link failure

Stage 1 (mistake)

$\downarrow$
gone

(Error propagation)

Hierar BT

$S_1 \rightarrow w \times (8) \quad O(\log N_t)$

$S_2 \rightarrow w \times (16)$

$S_3 \rightarrow w \times (32)$

$\log_2(1024) = 10 \text{ stages}$

faster

conv ... dependent bits $s_0 \rightarrow s_1$...

Beam Index $\rightarrow$ Digital msg

$$B_b = 110$$

$\downarrow$

conv Encoding

$\downarrow$

error correction

11 10 01 00 $\rightarrow$ code word

Encoded Bits

$\downarrow$

determine Beam patterns
tx during pilots

Rx sig strength

$\downarrow$
Beam alignment

CB $\rightarrow$ list of beam patterns

$w_1 \rightarrow$ Pattern A

$w_2 \rightarrow$ B

$w_3 \rightarrow$ C

$w_4 \rightarrow$ D

Indi Beam x

BS $\rightarrow$ Special combined beams

$$P_n = |y_n|^2$$

$$|h^H w_n + \text{noise}|^2$$

channel

$$y = h^H w_n + n$$

$\downarrow$
BF vector

How strong sig was for pilot beam

$P_1 \ P_2 \ P_3 \ \dots$

Beam Pattern
 $\downarrow$
Info about several directions at once

pilot $\rightarrow$ Info about real beam direction

Beam towards UE $\rightarrow h^H w_n$
 $\downarrow$
somewhere $\rightarrow$ small

Chi-Square Distribution

$y \rightarrow$ complex gaussian

$$|y|^2 = \text{real}^2 + \text{imag}^2$$

χ^2 distr $\rightarrow 2^\circ$ freedom

Beam $\rightarrow$ D

Non-central χ^2 distribution

$$y = 2+3j$$

$$P = |y|^2$$

$$y = 2+3j$$

$$P = 2^2 + 3^2 = 13$$

C1 Beam $\neq$ User

$$y \approx n$$

$P \approx |n|^2$ (central chi square dis)

orig $P \downarrow$

C2 Beam = U

$$y = \text{sig} + n$$

$$P = |\text{sig} + n|^2$$

non central χ^2 distribution

Ex:

WB

$$S=0$$

NP 0.3 0.5

CB

$$S=5$$

3, 3, 6, 1, 4, 8
moves right

Rx $\rightarrow$ Power

some beams

$\downarrow$
only noise

$\downarrow$
central χ^2

to decide which beam direction matches user channel

shifted away from 0

Viterbi Decoder

conv encoding $\rightarrow$ dependent bits

$R_x \rightarrow P_1 \ P_2 \ P_3 \ \dots$

$\downarrow$
LLR values (log likelihood ratio)

LLR1, LLR2, LLR3, $\dots$

$\downarrow$
encoded bit
1 (or) 0

2.1, -1.7, 0.8, -0.3
 $\downarrow$
Bit 1, 0, 1, uncertain

(Soft Int)

101 $\rightarrow$ 11 01 10
even wrong
we can recover

viterbi algo

Viterbi Algo

Searches (most likely bit seq)

Encoder $\rightarrow$ state machine

every path $\rightarrow$ calculate path metric = LLR
 $\hookrightarrow$ how well they match

State 00 $\rightarrow$ encoder

01 $\rightarrow$ trellis diagram
10
11

time $\rightarrow S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow$
 $S_1 \rightarrow S_2 \rightarrow S_3 \rightarrow$
 $S_2 \rightarrow S_3 \rightarrow S_4 \rightarrow$

110 $\xrightarrow{R_x}$ 11 10 01 $\rightarrow$ 10 10 00 $\xrightarrow{Upper}$ 110 $\xrightarrow{Decode}$ 6 $\xrightarrow{Best Beam}$ Beam 6

Setup encoders

convolutional encoder

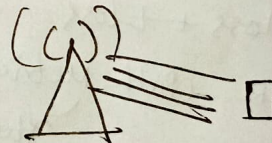
101 $\rightarrow$ 11 01 00 (More b $\rightarrow$ More prot)

trellis state

$$2^{(N-1)k}$$

$$(n, k, N) = (2, 1, 3) \rightarrow 2^2 = 4 \quad R = 1/2$$

training hierarchy tra



BS $\checkmark \rightarrow$ Beam?

8 steps $\rightarrow$ 3 steps

UE slg strong LB (or) RB

$\frac{B_0 \ B_1 \ B_2 \ B_3 \ B_4 \ B_5 \ B_6 \ B_7}{L \quad R \checkmark}$

$\frac{B_4 \ B_5 \ B_6 \ B_7}{\checkmark}$

$\frac{B_6 \ B_7}{\checkmark}$

$$y = h^T w + n \quad P = |y|^2$$

$$P = |h^T w + n|^2$$

$$P_A = |h^T w_A + n|^2$$

$$P_B = |h^T w_B + n|^2$$

P_C, P_D

$B_A \rightarrow 1 \text{ to } 4$
 $B_B \rightarrow 5 - 8$

$\frac{L_2 \ A(4)}{C \rightarrow 1-2 \quad D \rightarrow 3-4}$
 $\checkmark \frac{A \ B}{\checkmark}$

Beamforming Vector

$$w(\theta) = \frac{1}{\sqrt{Nt}} \begin{bmatrix} e^{j\pi\theta} \\ e^{j2\pi\theta} \\ \vdots \\ e^{j(Nt-1)\pi\theta} \end{bmatrix}$$

$$Nt = 4$$

B₁ B₂ B₃ B₄

$$h = \begin{bmatrix} B_1 \\ B_2 \\ B_3 \\ B_4 \end{bmatrix} \begin{bmatrix} 1 \\ 0.8 \\ 0.5 \\ 0.2 \end{bmatrix} \quad \begin{array}{l} P_A = 2.8 \\ P_B = 1.1 \\ P_A > P_B \end{array}$$

$$P = 3.0 > 2.0 \quad \text{errors}$$

training-hierarchy-softconv1

$$y = h^T w + n \quad \rightarrow \text{bf vector } P = |y|^2$$

$$\sigma = \sqrt{\frac{1}{\text{SNR}}} \quad Nt \rightarrow \text{directional beams}$$

Beam Codebook construct

$$w(\theta) = \frac{1}{\sqrt{Nt}} \begin{bmatrix} e^{j\pi\theta} \\ e^{j2\pi\theta} \\ \vdots \\ e^{j(Nt-1)\pi\theta} \end{bmatrix}$$

Beam In $\rightarrow$ conv code

$$\theta \in \left[-1 + \frac{1}{Nt}, 1 - \frac{1}{Nt}\right]$$

$$CR = K \ln = 112$$

$$g_1 = [1 \ 1 \ 1] \quad g_2 = [1 \ 0 \ 1]$$

$$11P \Rightarrow 101 \Rightarrow \text{Encode} \rightarrow \underline{11} \ 01 \ 00$$

BD

$$\text{states} = 2^{(N-1)K}$$

$$2^2 = 4$$

$$y_i = h^T w_i n$$

$$A_i = |y_i|^2$$

$$A = \text{abs}(y_i)^2$$

Viterbi Decoder

$\rightarrow$ to find most likely tx

bit seq

$$\text{loss} = -LLR$$

soft

$$LLR = \log \left(\frac{I_0(a)}{I_0(b)} \right) - \frac{\text{amp}^2}{\sigma^2} \quad (\text{Beampower} \rightarrow \text{prop info})$$

Bessel fun

State

$$\text{new-loss} = \text{old-loss} + t\text{-loss}$$

$$t\text{-loss} = -LLR \quad (\text{Path Metric}) \rightarrow \text{state}$$

$$n > \text{losses}$$

$$P_A \rightarrow 001$$

Path Metric

$\rightarrow$ How well path matches Rx sig

$$P_B \rightarrow 011$$

$$P_C \rightarrow 100$$

$$P_D \rightarrow 110$$

$$A = 8$$

$$(B \ 3)$$

$$\text{Min Ex: } N=3$$

$$K=1 \quad n=2$$

$$m=N-1$$

$$\text{Mem} = 1$$

$$\rightarrow \text{cur} + \text{pr}$$

$$\frac{(N-1)K}{2} = \frac{2}{2} = 1$$

State 0,1

every state depends on previous bit

$$A(1,1)$$

$$(1,2)$$

$$(1,3)$$

$$(1,4)$$

power $\rightarrow$

conf val of LLR

$$\text{llr}_1 \text{ llr}_2$$

Trellis

QIP

$S_0 \rightarrow 0$ (0,0)
 $S_0 \rightarrow 1$ (1,1)
 $S_1 \rightarrow 0$ (1,0)
 $S_1 \rightarrow 1$ (1,1)

every step produces 2 bits

Soft Info

S_1 LLR1 = 2
 LLR2 = 1
 S_2 LLR1 = -1
 LLR2 = 2

Bit
 -ve $\rightarrow 0$
 +ve $\rightarrow 1$
 S_1 $S_0 = 0$
 $S_1 = 0$

b-M (t-loss = -LLR)

T1 $S_0 \rightarrow S_0 \rightarrow (0,0)$

Metric loss $\Rightarrow -(-2) + -(-1) \Rightarrow -3$ (Penalty)

Path Metric $0 + (-3) = -3$

T2 $S_0 \rightarrow S_1 (1,1)$

Metric loss $= -(-2) + -(-1) = 3$

Path me $0 + 3 = 3$

(-3, 3)

Step 2

Step 1

State PM
 $S_0 \rightarrow -3$
 $S_1 \rightarrow 3$

S_2 LLR = -1, LLR = 2

(PA) $S_0 \rightarrow S_0 \rightarrow 00$
 $-(-1) + -(+2) = 1 - 2 = -1$

Metric $-3 + (-1) = -4$

$S_0 \rightarrow S_1 \rightarrow 11$

(P0)

Pen $= -(-1) + -(+2)$
 $1 - 2 = -1$

Met

$3 + (-1) = 2$

(P3) $S_0 \rightarrow S_1 \rightarrow 11$

Pen $-(-1) + -(+2)$
 $= 1 - 2 = -1$

Met $-3 + (-1) = -4$

(Pc) $S_1 \rightarrow S_0 (1,0)$

Pen $-(-1) + -(+2) = -3$

Met $3 + (-3) = 0$

Best Metric

$S_0 \rightarrow -4$
 $S_1 \rightarrow -4$

training hierarchy repeat

$L_1 = \log_2(Nt)$

$Nt = 8$

$L_1 = \log_2(8) = 3$

Noise

$\sigma^2 = 1/8NR$

Complex gaussian

CRS (steering vector)

$y = Hx^T w + n$

$P = |y|^2$

$B_1 \rightarrow 1$ to 4
 $B_2 \rightarrow 5$ to 8

L3

$B_1 \rightarrow 1$
 $B_2 \rightarrow 2$

L2

$B_1 \rightarrow 1-2$
 $B_2 \rightarrow 3-4$

Power

$A(1,1) \geq A(1,2)$

$id_{BS} = id_{BS}^* 2 + 0$

else

$id_{BS} = id_{BS}^* 2 + 1$

select beam compute gain

$A_1 \geq A_2$

$b = 0$

else $b = 1$

$id = 2id + b$, update index

Same → train-hic-gau

Noise $y = H^T w + n$ (complex gaussian)

Codebook

$n, k, N, m \rightarrow$ stages

HBT (iterations) $S \dots S_k$

UR → Path → $B_1 \rightarrow B_S \rightarrow$ Array gain

train-hic-tra

returns best B_1 & array gain

train-hic-softconv

uses soft decoding viterbi (conv code)

train-hic-soft

Codebook

↳ to select beams

Channel

↓

Codebook

↓

Beam Search → Soft Info

$S \dots S_k$

$y = H^T w + n$

Power

UR → decode

↓

$B_1 \rightarrow B_S \rightarrow$ Array gain

(power)

train-hic-repe

Multi-user

$B_0 \dots B_7 \rightarrow$ (Best Beam) (less levels)

train-hic-gau

(gaussian approximation)

UR, soft decoding (viterbi)

Best $B_S \rightarrow$ Array gain

train-exhaustive

best beam

check every possible beam

train-hic-conv

Hard decision

Min(B_S) X

Soft UR

hard decoding

↓

Best B_1

↳ array gain